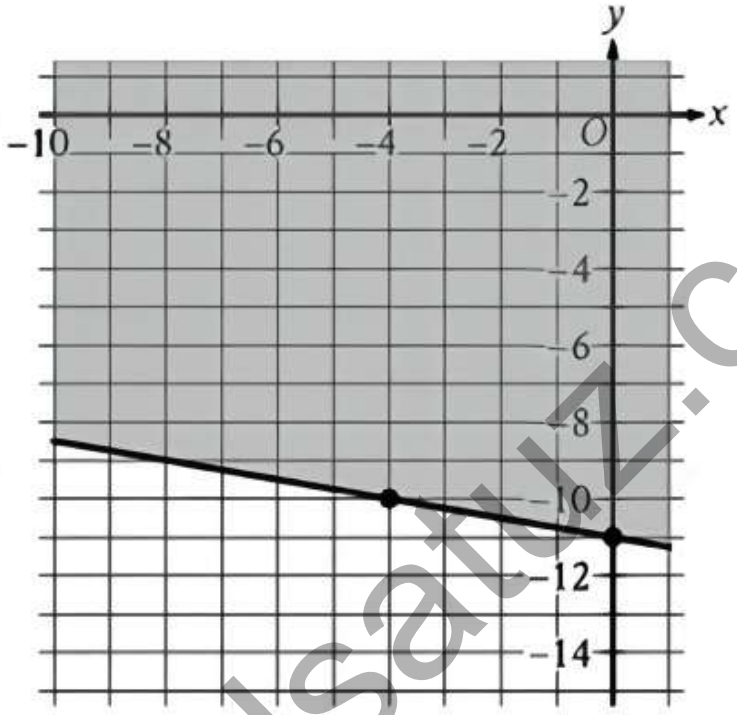


The shaded region shown represents the solutions to $rx + ty \geq -44$, where r and t are constants. What is the value of $r + t$?

(A) -4

(B) -3

(C) 3

(D) 5

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22 Mark for Review

In the xy-plane, line k and line ℓ are perpendicular and intersect at the point $(2, 8)$. If line k is defined by the equation $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on line ℓ?

(A) $(3, 8 - \frac{1}{m})$

(B) $(3, 8 + \frac{1}{m})$

(C) $(3, 8 - m)$

(D) $(3, 8 + m)$

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16 Mark for Review

A painting studio is offering a promotion where the first class is free, the second class is half off the regular price, and the remaining classes are regularly priced. If the regular price of a class is $\$26.40$, which function f gives the total cost, in dollars, of x classes taken using this promotion, where $x \geq 2$?

(A) $f(x) = 26.40(x - 1) + 13.20$

(B) $f(x) = 26.40(x - 2) + 13.20$

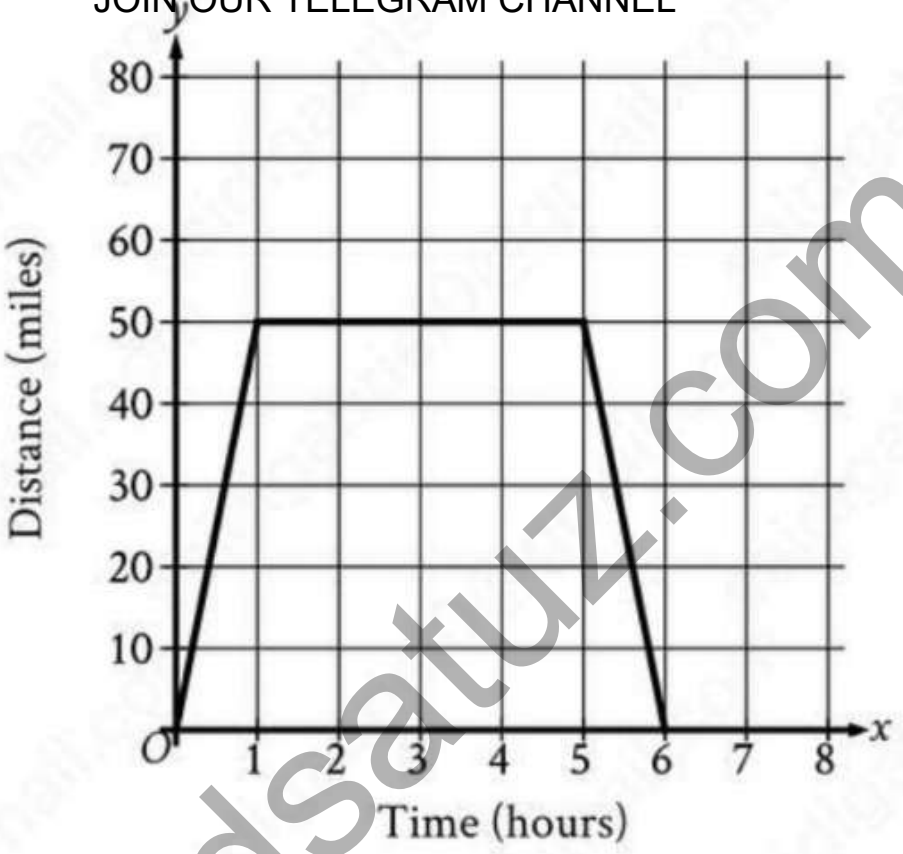
(C) $f(x) = 26.40(x - 1) + 13.20(x - 2)$

(D) $f(x) = 26.40(x - 2) + 13.20(x - 1)$

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Line s is defined by $x - 4y = 24$. Line t is perpendicular to line s in the xy-plane and passes through the points $(k, 0)$ and $(60, -k)$. What is the value of k?

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Quinidra drove her car on a straight road to a certain location. The graph below models Quinidra's distance, in miles, from her home as a function of time, in hours. Which of the following could the model describe?

(A) Quinidra drove away from her home at a constant speed for 1 hour, spent 4 hours at the location, then continued to drive away from her home at a constant speed for 1 hour.

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B Quinidra drove away from her home at a constant speed for 1 hour, spent 4 hours at the location, then drove back to her home at a constant speed for 1 hour.

C Quinidra drove away from her home at an increasing speed for 1 hour, spent 4 hours at the location, then drove back to her home at a decreasing speed for 1 hour.

D Quinidra drove away from her home at an increasing speed for 1 hour, drove at a constant speed for 4 hours, then drove back to her home at a decreasing speed for 1 hour.

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12 Mark for Review

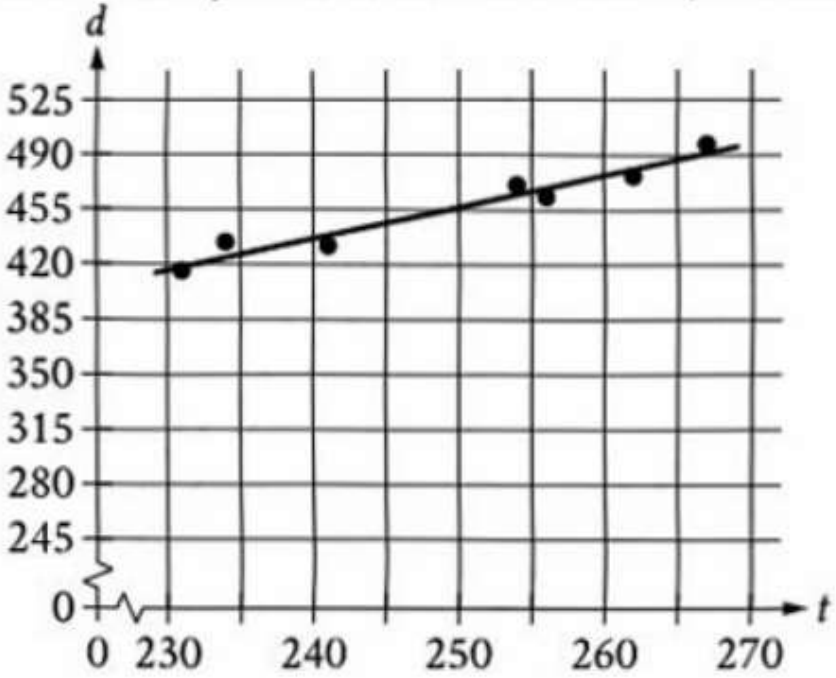
Line s in the xy-plane is defined by $6x + 2y = 0$. Line t is perpendicular to line s and passes through the points $(7, 0)$ and $(k, \frac{4}{3})$. What is the value of k?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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The scatterplot shows the relationship between variables t and d.



Which of the following equations is the most appropriate linear model for the data shown?

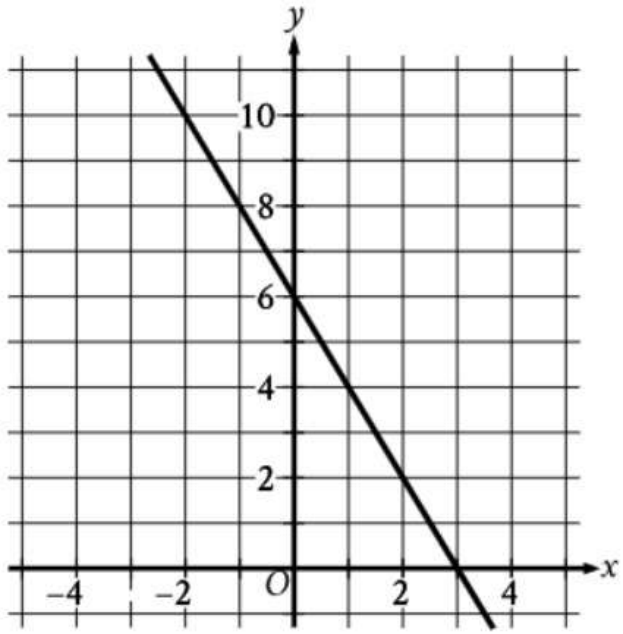
A. $d = -48.1 + 2.02t$

B. $d = 148.1 + 2.02t$

C. $d = 371.8 + 2.02t$

D. $d = 406.8 + 2.02t$

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The graph of the linear function $y = f(x) + 17$ is shown. If c and d are positive constants, which equation could define f?

A. $f(x) = -d - cx$

B. $f(x) = d + cx$

C. $f(x) = d - cx$

D. $f(x) = -d + cx$

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25 Mark for Review

$$f(x) = 3d(26x + 27) + 18$$

Which of the following represents the x-intercept of the graph of $y = f(x) + 6$ in the xy-plane, where d is a constant?

(A) $(81d + 24, 0)$

(B) $(\frac{-81d-24}{78d}, 0)$

(C) $(6 - \frac{81d+18}{78d}, 0)$

(D) $(-\frac{45}{78d+6}, 0)$

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